

Supplementary Table 3. Performance of the Waters and GMM functional classifiers.

Sensitivity, specificity, positive predictive value, and negative predictive value (each with a Wilson 95% confidence interval), together with the true- and false-positive and -negative counts (TP, FP, FN, TN), for the Waters classification (assigned from SGE q-values and directionality) and the GMM score-cutoff classification, against a clinical truth set (ClinVar pathogenic/likely-pathogenic vs benign/likely-benign) and a molecular truth set (truncating vs synonymous variants). All four metrics are computed only over variants each classifier called Abnormal or Normal; GMM's indeterminate calls (Indet. columns) are excluded from every denominator — e.g. GMM's specificity against the clinical truth set is 1,133/1,158, not 1,133/1,171.

Truth set	Method	TP	FP	FN	TN	Indet. (path)	Indet. (benign)	Sensitivity	Specificity	PPV	NPV
Clinical (ClinVar P/LP vs B/LB)	Waters	184	73	1	1098	0	0	0.995 (0.970–0.999)	0.938 (0.922–0.950)	0.716 (0.658–0.768)	0.999 (0.995–1.000)
Clinical (ClinVar P/LP vs B/LB)	GMM	184	25	1	1133	0	13	0.995 (0.970–0.999)	0.978 (0.968–0.985)	0.880 (0.829–0.918)	0.999 (0.995–1.000)
Molecular (Truncating vs Synonymous)	Waters	2797	123	56	1799	0	0	0.980 (0.975–0.985)	0.936 (0.924–0.946)	0.958 (0.950–0.965)	0.970 (0.961–0.977)
Molecular (Truncating vs Synonymous)	GMM	2779	51	64	1854	10	17	0.977 (0.971–0.982)	0.973 (0.965–0.980)	0.982 (0.976–0.986)	0.967 (0.958–0.974)